



DEPARTMENT OF ELECTRICAL ENGINEERING
UNIVERSITY OF SOUTH CAROLINA
COLUMBIA, SC 29208

Dear Barnes Scholarship Selection Committee,

Re: Jacob Vaught's application

I have known Jacob since Jan 2023 when he began working under a collaborative project between myself (materials scientist), Dr. David Matolak (signal processing), and Dr. Bridget Armstrong (public health) to build and test fitness trackers for children. Jacob has been primarily involved with signal processing to calibrate energy expenditure in children using accelerometry and heart rate signals from a USC built tracker. Jacob will further develop that tracker under an independent study this fall by translating it to printable flexible technology, techniques that are deeply relevant to the aerospace industry.

Jacob is an outstanding student, timely with communication, and has been enthusiastic with his contributions to this interdisciplinary project. I have primarily worked with him on Matlab scripts with a small portion of his effort on hardware debugging as he comes up to speed. He has an easy personality that lends him to working well in larger teams, and his communication skills, both written and verbal are excellent. In my relationship to him as a research advisor, I would place him in the top 5% of undergraduates at USC, a Carnegie R1 research university with the nation's premiere public Honor's College and hope to see him grow further in his career.

I am thrilled that Jacob is continuing to work for us this fall under the independent study on flexible electronics for wearables, and I intend to fund him so that he can focus on schoolwork. The Barnes scholarship will further help Jacob dedicate his energy completely to his research and engineering education. This will allow him to maximize his potential as an emerging scientist/engineer and contribute to our nation and society in areas of strategic importance. I recommend Jacob for this scholarship without reservation and expect that he will do great things. Feel free to contact me should you need any further information.

Sincerely,

MVS Chandrashekhar
Associate Professor
Electrical Engineering
607-435-5029 (cell)
chandra@sc.edu